PSTAT 5A F26

© Julie Swenson 2026

Discussion Section Worksheet 2

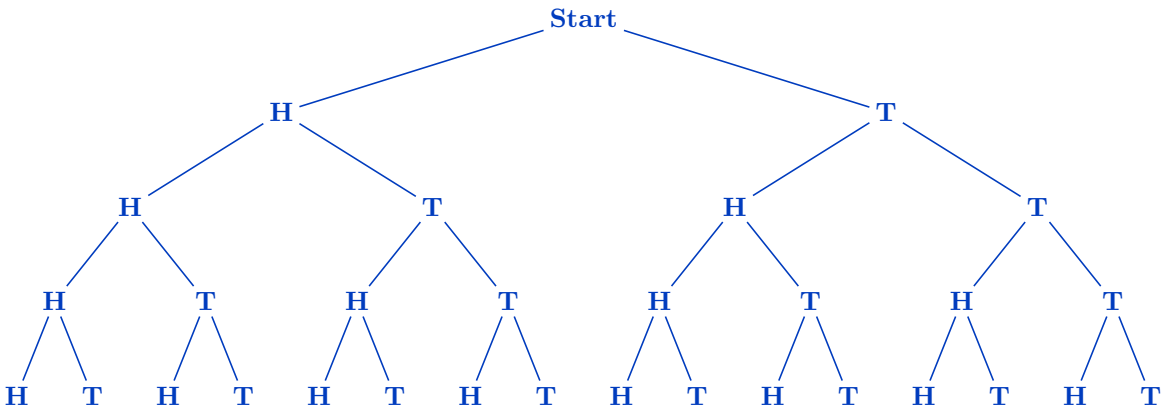
Solutions

Problem 1

1. A fair coin is flipped four times.

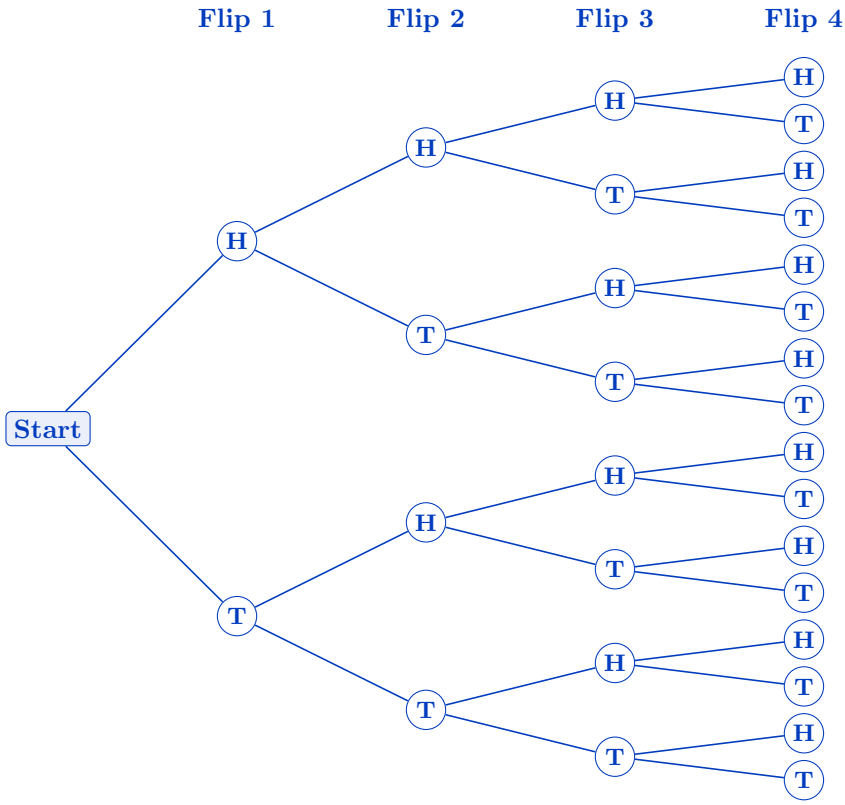
a) Draw a tree diagram for this random experiment.

Solution.



b) Find the sample space.

Solution. Each path of the tree gives one outcome; there are $2^4 = 16$ equally likely outcomes, each with probability $\frac{1}{16}$:



Outcome	No. of H
HHHH	4
HHHT	3
HHTH	3
HHTT	2
HTHH	3
HTHT	2
HTTH	2
HTTT	1
THHH	3
THHT	2
THTH	2
THTT	1
TTHH	2
TTHT	1
TTTH	1
TTTT	0

$$S = \left\{ \begin{array}{llll} \text{HHHH}, & \text{HHHT}, & \text{HHTH}, & \text{HHTT}, \\ \text{HTHH}, & \text{HTHT}, & \text{HTTH}, & \text{HTTT}, \\ \text{THHH}, & \text{THHT}, & \text{THTH}, & \text{THTT}, \\ \text{TTHH}, & \text{TTHT}, & \text{TTTH}, & \text{TTTT} \end{array} \right\}, \qquad |S| = 16.$$

c) We are interested in the number of heads observed. Define the random variable of interest, X.

Solution. X = the number of heads observed in four flips, $X \in \{0, 1, 2, 3, 4\}$.

d) Create a pdf for X.

Solution. Counting the outcomes in (b) by number of heads,

$$\mathrm{P}(X = x) = \frac{\#\{\text{outcomes with } x \text{ heads}\}}{16}.$$

From the diagram, the numbers of outcomes with $0, 1, 2, 3, 4$ heads are $1, 4, 6, 4, 1$.

x	0	1	2	3	4
$\mathrm{P}(X = x)$	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

e) Calculate the expected value of X.

Solution.

$$\mathrm{E}(X) = \sum_x x\,\mathrm{P}(X = x) = 0 \cdot \tfrac{1}{16} + 1 \cdot \tfrac{4}{16} + 2 \cdot \tfrac{6}{16} + 3 \cdot \tfrac{4}{16} + 4 \cdot \tfrac{1}{16} = \frac{32}{16} = 2.$$

Problem 2

2. How many textbooks do you need this quarter? Today, we study the distribution of textbooks needed (this quarter) by students in your section.

a) Define the random variable of interest, X.

Solution. X = the number of textbooks needed this quarter by a randomly selected student in the section.

b) Conduct a survey in your section. Write down the results.

Solution.

x	0	1	2	3	4	5	6	Total
Frequency $f(x)$	12	7	1	0	1	0	0	21

c) Use your results to create a probability distribution function (pdf).

Solution.

$$\mathrm{P}(X = x) = \frac{f(x)}{n} = \frac{f(x)}{21}.$$

x	0	1	2	3	4	5	6
$\mathrm{P}(X = x)$	$\frac{12}{21}$	$\frac{7}{21}$	$\frac{1}{21}$	0	$\frac{1}{21}$	0	0

Problem 3

3. A student is selected at random from your section. Find the probability that this student:

a) Needs 3 textbooks.

Solution.

$$\mathrm{P}(X = 3) = \frac{0}{21} = 0.$$

b) Needs at least 2 textbooks.

Solution.

$$\begin{aligned}\mathrm{P}(X \geq 2) &= \mathrm{P}(X = 2) + \mathrm{P}(X = 3) + \mathrm{P}(X = 4) + \mathrm{P}(X = 5) + \mathrm{P}(X = 6) \\ &= \frac{1 + 0 + 1 + 0 + 0}{21} = \frac{2}{21} \approx 0.0952.\end{aligned}$$

c) Does not need 5 textbooks.

Solution.

$$\mathrm{P}(X \neq 5) = 1 - \mathrm{P}(X = 5) = 1 - 0 = 1.$$

d) Needs between 2 and 4 textbooks.

Solution.

$$\mathrm{P}(2 \leq X \leq 4) = \mathrm{P}(X = 2) + \mathrm{P}(X = 3) + \mathrm{P}(X = 4) = \frac{1 + 0 + 1}{21} = \frac{2}{21} \approx 0.0952.$$

e) Needs at most 3 textbooks.

Solution.

$$\begin{aligned}\mathrm{P}(X \leq 3) &= \mathrm{P}(X = 0) + \mathrm{P}(X = 1) + \mathrm{P}(X = 2) + \mathrm{P}(X = 3) \\ &= \frac{12 + 7 + 1 + 0}{21} = \frac{20}{21} \approx 0.9524.\end{aligned}$$

f) Use the pdf to create a cumulative distribution function (cdf).

Solution.

$$F(x) = \mathrm{P}(X \le x) = \begin{cases} 0, & x < 0, \\ \frac{12}{21}, & 0 \le x < 1, \\ \frac{12+7}{21} = \frac{19}{21}, & 1 \le x < 2, \\ \frac{19+1}{21} = \frac{20}{21}, & 2 \le x < 4, \\ \frac{20+1}{21} = 1, & x \ge 4. \end{cases}$$

Problem 4

4. Use the data from your survey to calculate:

a) The mean of the distribution.

Solution. The mean of a discrete distribution is the weighted average of the possible values of X, where each value x is weighted by its probability $\mathrm{P}(X = x)$:

$$\mu = \mathrm{E}(X) = \sum_x x\,\mathrm{P}(X = x).$$

Using the pdf from Problem 2(c),

$$\begin{aligned} \mu &= 0 \cdot \tfrac{12}{21} + 1 \cdot \tfrac{7}{21} + 2 \cdot \tfrac{1}{21} + 3 \cdot 0 + 4 \cdot \tfrac{1}{21} + 5 \cdot 0 + 6 \cdot 0 \\ &= \frac{0+7+2+0+4+0+0}{21} = \frac{13}{21} \approx 0.6190. \end{aligned}$$

b) The variance of the distribution.

Solution. The variance measures how spread out the values of X are around the mean μ. It is the probability-weighted average of the squared deviations from the mean,

$$\sigma^2 = \mathrm{Var}(X) = \sum_x (x-\mu)^2\,\mathrm{P}(X = x),$$

which is algebraically equal to the shortcut formula

$$\sigma^2 = \mathrm{E}(X^2) - \mu^2, \qquad \text{where } \mathrm{E}(X^2) = \sum_x x^2\,\mathrm{P}(X = x).$$

Step 1: compute $\mathrm{E}(X^2)$. Square each value, then weight by its probability:

$$\begin{aligned} \mathrm{E}(X^2) &= 0^2 \cdot \tfrac{12}{21} + 1^2 \cdot \tfrac{7}{21} + 2^2 \cdot \tfrac{1}{21} + 3^2 \cdot 0 + 4^2 \cdot \tfrac{1}{21} + 5^2 \cdot 0 + 6^2 \cdot 0 \\ &= \frac{0+7+4+0+16+0+0}{21} = \frac{27}{21} = \frac{9}{7} \approx 1.2857. \end{aligned}$$

Step 2: subtract the squared mean. With $\mu = \frac{13}{21}$, $\mu^2 = \frac{169}{441}$, so

$$\sigma^2 = \frac{27}{21} - \frac{169}{441} = \frac{567}{441} - \frac{169}{441} = \frac{398}{441} \approx 0.9025.$$